Technology Stack

Project Title

Virtual Agent–Driven SLA Breach Awareness & Justification System

---

Introduction

The Technology Stack defines the software technologies, programming languages, frameworks, databases, and tools used to develop the **Virtual Agent–Driven SLA Breach Awareness & Justification System**. The selected technologies provide a scalable, secure, and efficient platform for proactive SLA monitoring, intelligent decision-making, workflow automation, and real-time reporting.

The project combines modern web technologies with Artificial Intelligence (AI) capabilities to improve IT Service Management (ITSM) by preventing SLA breaches before they occur and assisting users with timely recommendations and automated workflows.

Frontend Technologies

HTML5

HTML5 is used to design and structure the web pages of the application.

Features:

- Responsive page structure
- Forms for incident management
- Dashboard layout
- Virtual Agent interface
- Report pages

CSS3

CSS3 is used to improve the visual appearance and responsiveness of the application.

Features:

- Responsive design
- Modern user interface
- Dashboard styling
- Mobile-friendly layouts
- Interactive components

JavaScript

JavaScript provides dynamic functionality and improves user interaction.

Features:

- Form validation
- Dynamic dashboard updates
- API communication
- Interactive user interface
- Real-time data display

Chart.js

Chart.js is used to visualize SLA statistics and incident data.

Charts Used:

- SLA Status Distribution
- Incident Trends
- SLA Compliance Reports
- Ticket Priority Analysis
- Performance Metrics

Backend Technologies

Python

Python is the primary programming language used to develop the backend logic of the application.

Features:

- Easy integration with AI services
- Fast development
- High readability
- Extensive library support
- Secure backend processing

Flask

Flask is used as the backend web framework.

Responsibilities:

- Handle HTTP requests
- Manage application routes
- Process incident data
- Connect with the database
- Integrate Google Gemini AI
- Serve dashboard data

Database Technology

SQLite

SQLite is used as the database management system.

Stores:

- User information
- Incident records
- SLA details
- Breach justifications
- Notification history
- Dashboard statistics

Advantages:

- Lightweight
- Fast
- Easy to configure
- No separate server required
- Suitable for academic and prototype applications

Artificial Intelligence

Google Gemini AI

Google Gemini AI is integrated into the Virtual Agent to generate intelligent SLA breach explanations and corrective guidance.

Capabilities:

- Generate SLA breach justifications
- Explain SLA risks

- Recommend corrective actions
- Improve decision-making
- Assist support agents during incident handling

Service Management Platform

ServiceNow

ServiceNow provides the IT Service Management (ITSM) platform for incident and SLA management.

Modules Used:

- Incident Management
- Service Portal
- Virtual Agent
- SLA Management
- Reports and Dashboards
- User Administration

Workflow Automation

Flow Designer

Flow Designer automates SLA-related workflows.

Functions:

- Monitor SLA thresholds
- Trigger notifications
- Send email alerts
- Execute automated workflows
- Improve operational efficiency

Security

Role-Based Access Control (RBAC)

RBAC ensures that users can only access authorized information.

Roles:

- Administrator
- Team Lead / Manager

- Support Agent

Benefits:

- Secure data access
- Better accountability
- Controlled permissions
- Improved compliance

Development Tools

The following tools were used during development:

- Visual Studio Code
- Python
- Flask Framework
- SQLite
- Google Gemini AI
- ServiceNow Platform
- Flow Designer
- Git
- GitHub
- Google Chrome

System Architecture Overview

The application follows a **three-tier architecture**.

Presentation Layer

- HTML5
- CSS3
- JavaScript
- Chart.js

Application Layer

- Python
- Flask
- Business Logic

- Virtual Agent
- Google Gemini AI
- Flow Designer

Data Layer

- SQLite Database

Advantages of the Selected Technology Stack

- Lightweight architecture
- Easy maintenance
- Secure user management
- AI-assisted decision support
- Real-time SLA monitoring
- Automated workflows
- Fast application performance
- Scalable design
- Interactive dashboards
- Efficient data management

Conclusion

The selected technology stack provides a robust foundation for developing the **Virtual Agent–Driven SLA Breach Awareness & Justification System**. The combination of **Python, Flask, HTML5, CSS3, JavaScript, Chart.js, SQLite, ServiceNow, Flow Designer, and Google Gemini AI** enables the application to proactively monitor SLA performance, automate incident workflows, generate intelligent recommendations, and deliver a secure, scalable, and user-friendly IT Service Management solution.